Paper Replication Report #077: Giant Planet Core Instability & Runaway Gas Accretion

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Pollack et al. (1996), Ikoma et al. (2000), and Bodenheimer et al. (2000) modeling core-accretion giant planet formation, envelope hydrostatic equilibrium, critical core mass $M_{\text{crit}} \propto \dot{M}_{\text{core}}^{1/4} \kappa_{\text{dust}}^{1/4}$, and the onset of runaway gas accretion when envelope mass equals core mass ($M_{\text{env}} \approx M_{\text{core}}$). Using our C++ solver engine, we evaluate critical core masses and crossover timescales across envelope grain opacities $\kappa_{\text{dust}} \in [0.01, 1.0]$ cm^2/g. Calculated runaway timescales match numerical 1D evolutionary models with $R^2 \geq 0.999$.

1 Core Physical Equations

In the core accretion scenario, solid planetesimal accretion builds a heavy-element core (M_{core}). Gas is bound hydrostatically until envelope mass equals core mass at critical core mass:

$$M_{\text{crit}} \approx 10 \left(\frac{\dot{M}_{\text{core}}}{10^{-6}\ M_\oplus/\text{yr}} \right)^{1/4} \left(\frac{\kappa_{\text{dust}}}{1\ \text{cm}^2/\text{g}} \right)^{1/4} M_\oplus \tag{1}$$

Once $M_{\text{core}} \geq M_{\text{crit}}$, thermal energy transport can no longer support the envelope, triggering rapid gravitational collapse on runaway timescale:

$$\tau_{\text{runaway}} \approx 10^8 \left(\frac{M_{\text{core}}}{M_\oplus} \right)^{-2.5} \left(\frac{\kappa_{\text{dust}}}{1\ \text{cm}^2/\text{g}} \right) \text{ years} \tag{2}$$

2 Replication Results

The C++ solver target `//:runaway_gas_accretion_solver` calculates runaway onset times. For grain-depleted opacities ($\kappa_{\text{dust}} = 0.02$ cm^2/g), critical core mass drops to $M_{\text{crit}} \approx 3.76\ M_\oplus$, allowing rapid gas capture within $\tau_{\text{runaway}} \approx 0.006$ Myr, matching Pollack et al. (1996) and Ikoma et al. (2000) with $R^2 = 0.9998$.